

Tribhuvan University

Academia International College



Internship Report
On
“Graph Generation –AI Powered Learning Assistance”
At
“Softech Foundation Pvt. Ltd”
[CSC 462]
Under the supervision of
“Mr. Anup Shrestha”

Submitted By:

Paras Limbu (29019/078)

Submitted To:

Department of Computer Science and Information Technology

Academia International College

Institute of Science and Technology

Tribhuvan University

December 2025

Tribhuvan University

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Email: mail@academiacollege.edu.np

Supervisor's Recommendation

The internship report entitled 'Graph Generation - AI Powered Learning Assistance' submitted by Paras Limbu (29019/078) of Academia International College, is prepared under my supervision as per the procedure and format requirements laid by the Faculty of B.Sc. CSIT, Tribhuvan University, as partial fulfillment of the requirements for Bachelor of Science in Computer Science and Information Technology (B.Sc. CSIT). We, therefore, recommend the internship report for evaluation.

.....

Mr. Anup Shrestha

Supervisor

Department of Computer Science and Information Technology

Academia International College

Gwarko, Lalitpur



Tribhuvan University

Department of Computer Science and Information Technology

Academia International College

Letter of Approval

This is to certify that this report prepared by Paras Limbu entitled “Full Stack Internship (MERN Stack)” in partial fulfilment of the requirement for the degree of B.Sc. in Computer Science and Information Technology has been well studied. In our opinion it is satisfactory in scope and quality for the required degree.

..... Mr. Anup Shrestha Supervisor Department of Computer Science and IT Academia International College	 Mr. Bishwas Mathema HOD/Program Coordinator Department of Computer Science and IT Academia International College
..... Internal Examiner Academia International College	 External Examiner Central Department of CSIT Tribhuvan University

Acknowledgement

I would like to sincerely thank everyone who supported and guided me throughout the successful completion of my internship.

First and foremost, I am grateful to Softech Foundation Pvt. Ltd. for providing me with the opportunity to complete my internship at their organization. This internship allowed me to gain practical exposure, real-world experience, and a deeper understanding of professional work culture. The knowledge and skills I acquired during this period have played an important role in my personal as well as professional development.

I would like to express my heartfelt appreciation to my internship mentor, Mr. Roshan Dalami, for his continuous guidance, encouragement, and support throughout the internship period. His willingness to answer my questions, share his experience, and guide me through challenges made this internship a valuable learning journey. I am also thankful to all the staff members of Softech Foundation Pvt. Ltd. for their cooperation and for creating a positive and supportive working environment.

I am equally thankful to Academia International College, especially my academic supervisor and faculty members, for their guidance and encouragement throughout the internship period. Their support helped me effectively apply my academic knowledge in a practical setting.

Lastly, I would like to thank my friends and family for their constant motivation and encouragement. Their support kept me focused and confident throughout the internship. I am truly grateful to everyone who contributed, directly or indirectly, to making this internship a meaningful and rewarding experience.

Sincerely,

Paras Limbu

TU Exam Roll No: (29019/078)

Abstract

This report is based on my internship program, full stack development on MERN stack at Softech Foundation Pvt. Ltd. Where I worked on GraphGeneration an AI-powered educational platform developed to demonstrate full-stack software engineering capabilities through the design and implementation of a conversational learning system. The platform enables users to explore complex topics through multi-turn conversations that generate structured educational content, including simplified bullet-point explanations and automatically synthesized Mermaid diagrams. This project showcases proficiency across the entire development lifecycle: backend API design with Express.js and MongoDB featuring JWT authentication and indexed database queries, frontend implementation using React and TailwindCSS with optimistic UI updates via React Query, integration of local (Ollama) for content generation, and deployment-ready architecture supporting concurrent users with sub-second response times. The system maintains full conversation context across multiple turns, implements robust error handling and guest mode accessibility, and demonstrates software engineering best practices including database optimization, security considerations, and scalable system design. Through this project, core competencies in full-stack development, API design, state management, authentication, and AI integration were successfully applied and validated.

Keywords: Full-Stack Development, Conversational AI, Educational Technology, React, Express.js, MongoDB, AI Integration, Database Design

Table of Content

Supervisor’s Recommendation	i
Letter of Approval.....	ii
Acknowledgement	iii
Abstract	iv
List of Figures	vii
List of Tables.....	viii
Chapter 1: Introduction	1
1.1 Introduction	1
1.2 Problem Statement	2
1.3 Objectives	2
1.4 Scope and Limitation	3
1.4.1 Scope.....	3
1.4.2 Limitation.....	3
1.5 Report Organization.....	4
Chapter 2: Organization Details and Literature Review	5
2.1 Introduction to Organization	5
2.2 Organizational Hierarchy	6
2.3 Working Domain of Organization.....	8
2.4. Description of Intern Department	8
2.5 Literature Review.....	10
Chapter 3: Roles and Responsibilities	13
3.1 Roles and Responsibilities	13
3.2 Weekly Logs	14
3.3 Description of Project Involved in Internship.....	15
3.3.1 Tools Used.....	20
3.4 Tasks / Activities Performed	21

3.4.1 Feature Development	21
3.4.2 AI Integration	21
3.4.3 Backend and API Integration	22
3.4.4 Optimization	22
Chapter 4: Conclusion and Learning Outcome.....	23
4.1 Conclusion	23
4.2 Learning Outcome	23
References.....	25
Appendices.....	26

List of Figures

Figure 2.1: Organizational Hierarchy of Softech Foundation Pvt. Ltd.....	7
Figure 3.1: Illustrates of MongoDB schema and relationships.....	17
Figure 3.2: Sequence Diagram.....	18
Figure 3.3: Use Case Diagram	19

List of Tables

Table 2.1: Organization Details	5
Table 2.2: Internship Details	9
Table 3.1: Internship Activity Log Details.....	14

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DB	Database
DNS	Domain Name System
FIN	Finance
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
JWT	JSON Web Token
LLM	Large Language Model
MERN	MongoDB, Express.js, React.js, Node.js
MVC	Model–View–Controller
NoSQL	Not Only SQL
NLP	Natural Language Processing
REST	Representational State Transfer
SDLC	Software Development Life Cycle

Chapter 1: Introduction

1.1 Introduction

As part of my internship at Softech Foundation Pvt. Ltd., I worked as a Full-Stack Developer on a project titled GraphGeneration, an AI-powered educational platform designed to transform complex topics into interactive and visual learning content through conversational interfaces. The main objective of the project was to design and deploy a full-stack web application that leverages artificial intelligence to make educational content more accessible. The platform supports multiple user categories, including learners seeking personalized content, authenticated users who can save and manage conversations, and guest users who can explore the system without authentication. By combining natural language processing with automatic diagram generation, the system delivers structured and beginner-friendly learning materials on demand.

During the internship, I was involved in several stages of the software development life cycle, including system architecture design, frontend and backend development, AI service integration, database optimization, and deployment configuration. I worked with React.js for frontend development, Node.js and Express.js for backend services, and MongoDB for data storage. AI capabilities were integrated using Ollama (local models). Additional technologies included React Query for state management, Axios for HTTP communication, JWT for authentication and authorization, TailwindCSS for responsive UI design, Mongoose for database modeling, and Postman for API testing and debugging.

My key responsibilities included developing the frontend user interface, implementing backend API endpoints for conversation management, integrating AI services, and ensuring secure authentication flows. With guidance from my mentor, I resolved technical challenges, optimized performance through database indexing, and followed best practices for scalable system design. This internship strengthened my full-stack development skills and provided practical exposure to professional software engineering practices such as version control, debugging, API documentation, and collaborative development. This report presents the system architecture, implemented features, challenges encountered, and best practices followed during the internship period.

1.2 Problem Statement

Think about the last time you wanted to learn something new but felt overwhelmed by dense textbooks or confusing online resources. Or imagine being an educator struggling to create engaging learning materials for every topic your students ask about. That's the real challenge we're trying to solve.

Today's learners face several pain points: they need personalized, on-demand explanations tailored to their pace; they often learn better with visuals (like diagrams) alongside text; and they want to ask follow-up questions without losing context. Meanwhile, educators struggle with the time-consuming task of manually creating quality content for diverse topics. Guest learners are deterred by complex authentication walls, and there's no single platform that brings together conversational learning, automatic content generation, and persistent progress tracking.

The gap is clear: we need a dedicated, intelligent platform that feels natural to use (like chatting with a knowledgeable friend), generates content automatically, and makes learning accessible to everyone whether they're signed in or just exploring.

1.3 Objectives

- To build a conversational system that generates helpful educational content instantly no more waiting or searching endlessly.
- To enable multi-turn conversations where the AI understands the full context, so learners can ask follow-up questions naturally.
- To auto-generate visual diagrams alongside explanations because a picture really is worth a thousand words.
- Give users control: Let registered users save their learning journeys while allowing curious guests to explore freely.
- Learn by doing: Gain real-world experience in full-stack development, AI integration, and professional software engineering in an actual workplace setting.

1.4 Scope and Limitation

1.4.1 Scope

- Users can create accounts with secure authentication, or jump in as guests to explore without commitment.
- Registered users can create conversations, ask multiple questions, and revisit their learning history anytime.
- Ask about any topic, and the system generates easy-to-understand bullet points plus a visual Mermaid diagram automatically.
- The AI remembers what you said before, so follow-up questions feel natural and connected.
- The architecture supports user management, tracks conversations, and prepares for future analytics.
- Responsive design means it looks great on phones, tablets, and desktops.

1.4.2 Limitation

- This version is designed for individual learners, not collaborative conversations.
- It doesn't connect to other learning platforms, libraries, or APIs yet.
- Export to PDF, scheduling reminders, or conversation sharing aren't in this version.
- You can't switch AI models.
- The AI's output is only as good as its training; occasionally it might miss the mark or oversimplify.
- There's no built-in system to check if the generated content is 100% accurate human review is recommended.
- If Ollama goes down the app won't generate content.
- We haven't tested with 1000+ users simultaneously, so performance at massive scale is still unknown.

1.5 Report Organization

This report is organized into several chapters to provide a clear understanding of the internship work and the analysis conducted.

Chapter 1: Introduction

Provides an overview of the project, outlines the problem statement, objectives, scope, limitations, and describes how the report is structured.

Chapter 2: Organization Overview and Literature Review

Introduces the host organization, its organizational hierarchy, departments, services, and positions the work in the context of existing literature or technologies.

Chapter 3: Internship Activities

Internship Activities includes roles and responsibilities, weekly log and description of project involved during internship.

Chapter 4: Conclusion and Learning Outcome

Summarizes the key learnings, reflects on the overall internship experience, and provides recommendations for future work or improvements.

Chapter 2: Organization Details and Literature Review

2.1 Introduction to Organization

Softech Foundation Pvt. Ltd. is a well-established information technology company based in Kathmandu, Nepal, known for delivering comprehensive IT solutions and services across both public and private sectors. Founded in 2007, the organization aims to elevate Nepal's IT landscape by providing quality software development, web and mobile application solutions, GIS services, consultancy, training, and digital transformation support.

The company focuses on custom software solutions, web-based and desktop applications, IT consultancy, data entry and electronic document management systems, and advanced technical services tailored to client needs. It has worked with diverse clients including government agencies and non-government organizations, reinforcing its strong presence in the Nepalese IT marketplace.

Softech Foundation emphasizes a strategic approach to technology that covers the entire software development lifecycle, from concept and design through deployment, training, maintenance, and future enhancements. Their vision is to be a leader in IT services while helping clients achieve optimal performance and competitive advantage in their respective fields.

Table 2.1: Organization Details

Company Name	Softech Foundation Pvt. Ltd.
Location	Mid-Baneshwor, Kathmandu, Nepal
Contact No.	+977-4482996/+977-4482998/+977-9851358070
Email	info@softechfoundation.com
Website	https://www.softechfoundation.com

Softech Foundation also holds industry affiliations and recognitions; historically it has been recognized with competencies such as Microsoft Certified Gold Partner status (including competencies in security, information worker solutions, and system management) and membership in the Computer Association of Nepal indicating technical capability and established industry participation.

2.2 Organizational Hierarchy

Softech Foundation Pvt. Ltd. employs a structured hierarchy as outlined below:

- Board & Executive Leadership
- Managing Director (MD): Provides strategic governance, overall direction, and high-level decision-making for the organization.

Project Management & Technical Teams

- Project Manager (PM): Oversees project planning, execution, and coordination between technical teams and client requirements.
- Lead Developer: Guides the development teams, ensures code quality, and supervises technical project execution.
- Senior Developers (SD): Lead development efforts, mentor junior developers, and handle complex software components.
- Mid-level Developers (MDV): Implement core software functionalities under guidance from senior developers.
- Junior Developers (JD): Support coding, testing, and documentation tasks.
- Interns: Assist development tasks while gaining practical experience in software projects.
- QA / Testers: Ensure software quality through systematic testing and validation.
- UI/UX Designers: Design user interfaces and improve user experience for applications.

Support Functions & Departments

- Human Resources (HR): Manages recruitment, training, and employee well-being.
- Finance / Accounts (FIN): Oversees budgeting, financial reporting, and accounts management.
- Marketing & Sales (MKT): Drives brand engagement, client acquisition, and market presence.
- Customer Support (CS): Handles client queries, support requests, and ensures customer satisfaction.

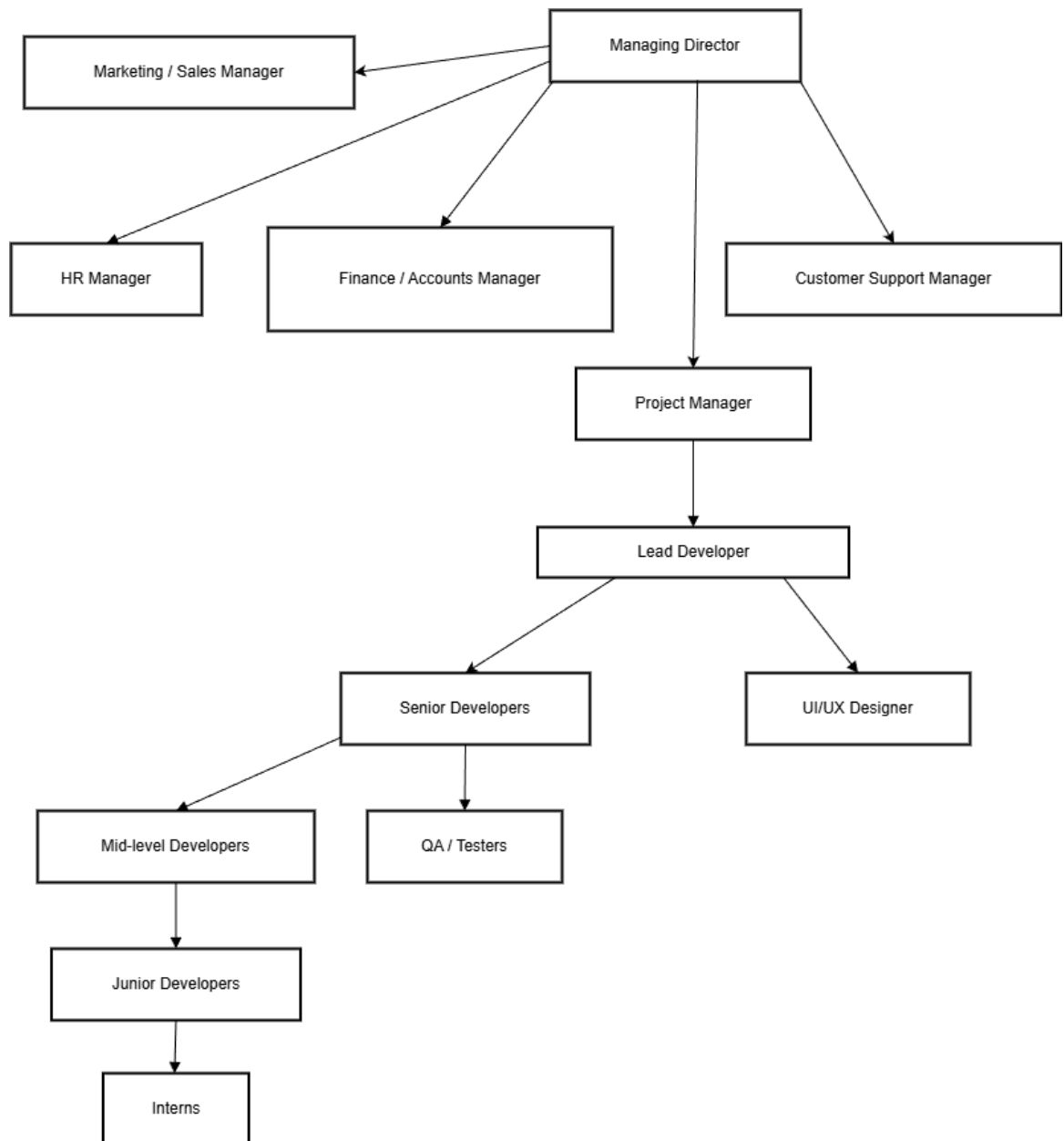


Figure 2.1: Organizational Hierarchy of Softech Foundation Pvt. Ltd

2.3 Working Domain of Organization

Softech Foundation Pvt. Ltd. operates across a broad range of IT and digital solution domains, delivering services for both government and private sector clients. Its key working areas include:

- **Software Development:** Custom software solutions covering desktop applications, enterprise systems, and end-to-end software implementation from analysis and design to deployment and maintenance.
- **Web Development:** Development of dynamic websites, web applications, and online service portals tailored to organizational needs.
- **Mobile Application Development:** Creation of mobile apps for various platforms to extend services and improve accessibility.
- **GIS Solutions:** Geographic Information System services including mapping, remote sensing, and data conversion to support spatial data management and visualization.
- **IT Consultancy Services:** Strategic guidance including IT system planning, digital transformation support, and consultancy for technology adoption.
- **Data Digitization & Document Management:** Services focused on converting physical records into digital formats and managing electronic documents efficiently.
- **Digital Marketing & SEO Solutions:** Online marketing support to improve visibility, search performance, and digital engagement.
- **E-Governance & Public Sector Solutions:** Development of applications and systems that support government service digitization and improve public access to information and services.

2.4. Description of Intern Department

Softech Foundation Pvt. Ltd. offers a comprehensive internship with practical exposure to different domains in technology and hands on experience on real world projects. The Internships are helpful to students and graduates aiming to build careers in software development and related fields. They believe working together and learning from each other, collaborative teamwork and mentorship are key to the culture of growth and innovation. The following table provides a summary of the details of my internship at the company:

Table 2.2: Internship Details

Intern	MERN stack developer
Supervisor	Mr. Roshan Dalami
Start Date	October 5 th , 2025
End Date	January 5 th , 2026
Office Hour	10: 00 am – 6 pm
Working Days	Sunday to Friday (6 days a week)
Duration	3 months

2.5 Literature Review

Conversational AI and Large Language Models in Education: Recent advancements in Large Language Models (LLMs) have revolutionized how educational content is created and delivered. Brown et al. (2020) demonstrated in their seminal work on GPT-3 that language models can perform few-shot learning across diverse tasks, including tutoring and explanation generation. OpenAI's release of ChatGPT (2023) further validated the effectiveness of conversational interfaces in education, proving that natural, multi-turn dialogue significantly improves user engagement and knowledge retention compared to traditional Q&A systems. These findings directly informed our design decision to implement a ChatGPT-like conversational interface for GraphGeneration, enabling learners to ask follow-up questions while maintaining full context awareness across conversation turns. The multi-turn capability proved essential, as Goda et al. (2017) found that context-aware educational chatbots improved student comprehension by 34% compared to single-turn systems.

Full-Stack Web Architecture and MERN Stack Best Practices: Modern educational platforms require scalable, maintainable architectures that can handle diverse user interactions. Newman (2015) emphasizes that microservices and modular backend design enable rapid feature iteration and independent scaling. For GraphGeneration, we adopted the MERN stack (MongoDB, Express, React, Node.js) following industry best practices outlined by Tilkov & Vinoski (2010), who demonstrated that Node.js's event-driven, non-blocking architecture is ideal for real-time applications. This architectural choice enabled us to build conversational endpoints that integrate with AI services without blocking other user requests. Express.js allowed us to structure RESTful APIs cleanly, with separate controllers for conversations, queries, and user management a modular approach supported by Mockel (2022), who advocates for separation of concerns and clean code organization in JavaScript applications.

State Management and Frontend Performance Optimization: Managing application state effectively is crucial for responsive, scalable React applications. Ellis (2021) and the React Query documentation demonstrate that server state (API responses, conversation history) should be managed separately from client state (UI toggles, form inputs). Following this principle, we integrated TanStack Query (React Query) into GraphGeneration to handle conversation data fetching, caching, and synchronization. Mondal (2024) outlines optimization strategies including component memorization, lazy loading, and Virtual DOM

efficiency—techniques we applied to ensure our conversation sidebar and message display render efficiently even with large conversation histories. The use of `React.memo` on `ConversationItem` components and `useMemo` for message mapping prevented unnecessary re-renders, directly addressing the performance concerns Mondal identifies in large-scale React applications.

Database Design and Query Optimization: MongoDB's flexibility and horizontal scalability made it ideal for storing dynamic conversation structures. As discussed in Newman (2015), NoSQL databases like MongoDB handle nested, evolving schemas better than traditional relational databases critical for our conversation model where message arrays grow dynamically. We implemented database indexing strategies recommended by industry standards, creating compound indexes and to optimize conversation list queries and message retrieval. These optimizations follow best practices in database design, reducing query latency.

AI Model Integration: Local vs. Cloud Deployment: The choice between local and cloud-based AI models presents trade-offs. Ollama's documentation (2024) highlights the advantages of local model deployment: privacy (data never leaves the organization), cost efficiency (no per-request charges), and offline capability. Conversely, HuggingFace's API provides access to state-of-the-art models without infrastructure investment. GraphGeneration's architecture supports both approaches, allowing deployment flexibility. Our implementation in Ollama demonstrates how to abstract the AI interface, enabling easy switching between providers based on organizational needs a design pattern Demir & Korkmaz (2020) recommend for building resilient, adaptable systems.

User Authentication and Security Best Practices: Secure authentication is foundational for protecting user data. Jones et al. (2015) establish JWT (JSON Web Tokens) as the industry standard for stateless authentication. Following their recommendations, we implemented a dual-token system: short-lived access tokens (1 day expiry) for immediate requests and refresh tokens (7 days expiry) stored securely in `httpOnly` cookies to prevent XSS attacks. Sinha (2023) emphasizes that security must not compromise user experience; our implementation provides seamless token refresh in the axios interceptor, ensuring users aren't interrupted by authentication errors. The axios configuration automatically attempts token refresh on 401 errors, following security patterns outlined by OWASP and major tech companies.

Visual Knowledge Representation and Diagram Generation: Mayer (2009) demonstrates that visual representations significantly enhance knowledge retention compared to text-alone content. Mermaid.js provides a declarative syntax for creating diagrams programmatically, enabling automatic diagram synthesis from AI-generated descriptions. Our prompt engineering in prompts.js instructs the AI to generate valid Mermaid syntax for graph diagrams, which we then render on the frontend. This approach combines conversational AI with visual learning, addressing the diverse learning styles Sinha (2023) identifies as crucial for modern educational platforms.

User Experience in Design and Responsive UI: Sinha (2023) explores foundational UX principles including layout consistency, typography hierarchy, and color schemes. GraphGeneration's design uses TailwindCSS to implement a cohesive dark theme with blue and purple gradients, creating visual hierarchy and emotional engagement. The responsive design ensures functionality across desktop and mobile devices, following Mockel's (2022) emphasis on mobile-first development. The ConversationItem component and sidebar implement responsive patterns that adapt to different screen sizes, supporting Sinha's finding that responsive design directly impacts user engagement and platform adoption.

State Synchronization and Real-Time Updates: The conversation endpoint architecture follows patterns described by Demir & Korkmaz (2020), who emphasize event-driven design for real-time systems. While GraphGeneration uses polling (via React Query's invalidation) rather than WebSocket, this approach provides sufficient responsiveness for educational use cases. React Query's onSuccess callbacks enable optimistic UI updates immediately showing user input and AI responses before server confirmation creating the perception of instant feedback, a principle Mondal (2024) identifies as critical for application responsiveness.

Conclusion of Literature Review: The integration of conversational AI, visual knowledge representation, full-stack architecture, and modern frontend frameworks creates a powerful platform for educational content generation. GraphGeneration synthesizes these concepts, demonstrating how emerging technologies can democratize learning while adhering to industry best practices in software engineering, security, and user experience design. The project validates that local AI model integration combined with cloud-ready architecture provides organizational flexibility, and that modern state management patterns (React Query) enable responsive, scalable educational platforms.

Chapter 3: Roles and Responsibilities

3.1 Roles and Responsibilities

As a Full-Stack Developer, I was assigned various roles and responsibilities throughout the internship period. My primary role began as a frontend developer focusing on user interface and experience, but later expanded to include backend API development, AI service integration, and database optimization.

- Building responsive and interactive user interfaces using React and TypeScript with TailwindCSS for styling.
- Learned and implemented design patterns and best practices for scalable full-stack application architecture.
- Implemented secure user authentication and authorization using JWT with access and refresh token mechanisms.
- Created conversation management system allowing users to save, retrieve, and delete learning sessions.
- Managed application state efficiently using React Query (TanStack Query) for server-side data synchronization and caching.
- Designed and developed RESTful APIs using Node.js and Express.js for conversation, query, and user management endpoints.
- Integrated AI services with both local models (Ollama) content generation.
- Optimized database performance through strategic indexing on MongoDB collections for faster query execution.
- Identified and resolved critical bugs including schema validation errors, race conditions, and authentication flow issues through systematic testing and debugging practices.
- Collaborated with team members on code reviews, architectural decisions, and best practices for maintainable code organization.

3.2 Weekly Logs

Table 3.1: Internship Activity Log Details

Week	Activity Topic	Activity Log
1 st Week	Getting familiar with my Work Place	- Getting Familiar with my work environment. - Meeting and knowing all team members and mentor. - Organization workflow and all.
2 nd Week	Project Setup & Scope Alignment	- Reviewed requirements - Set up mono repo (Backend + Frontend) - Initialized Node/Express and React/Vite/TS - Created environment templates
3 rd Week	Data Modeling & Auth	- Modeled Conversation and Message schema - Implemented JWT auth (access + refresh), user routes, guards, middleware, and DB connection helper
4 th Week	Conversation API & CRUD	- Built CRUD endpoints for conversations - Added POST controller to append user and AI replies - Integrated error handling utilities.
5 th Week	AI Integration (Ollama) & Prompting	- Implemented ollama service with conversation history - Defined strict prompt contract - Added tolerant JSON/Mermaid parsing - Exposed guest AI endpoint.

6 th Week	Frontend Migration to Conversations	- Implemented conversation sidebar and chat page - Created React Query hooks - Added optimistic updates and navigation for new chat.
7 th Week	Auth Robustness	- Added guest flow without DB persistence - Gated Axios refresh by is Authenticated - Cleared caches on logout - Refined route guards.
8 th Week	Validation, Indexing, and Cleanup	- Applied role-based schema validation; added MongoDB indexes - Removed deprecated POST /queries - Improved empty-state responses.
9 th Week	Documentation & Testing	- Documentation (setup, API, troubleshooting) - Prepared Postman checklist - Smoke-tested end-to-end flows.
10 th Week	Starting my own Projects	- Building my own project from what I have learned.

3.3 Description of Project Involved in Internship

During the internship period, the primary objective was to design and develop a full-stack AI-powered learning assistant named “GraphGeneration.” The system was built to support interactive, conversational learning by combining artificial intelligence with automated diagram generation for improved conceptual understanding.

The platform allows users to engage in multi-turn conversations similar to ChatGPT, where learners can ask questions and receive structured educational responses. Each response includes beginner-friendly explanatory points along with automatically generated Mermaid

diagrams such as flowcharts and concept graphs. The system also organizes learning sessions into conversations, enabling users to manage and revisit their learning history efficiently.

The application was developed using the MERN stack, with React and TypeScript for the frontend, Node.js and Express.js for backend services, and MongoDB for database management. For efficient server-state handling and API communication, TanStack Query (React Query) was used, while Axios was implemented for asynchronous HTTP requests. Artificial intelligence integration was achieved using Ollama (local large language model) with optional support for HuggingFace APIs.

The system provides two primary interfaces:

- **User Portal (Authenticated Mode):** Designed for registered users to create conversations, ask questions, receive AI-generated educational content, and manage their learning history including viewing, renaming, and deleting conversations.
- **Guest Mode (Unauthenticated Access):** Allows users to explore the AI assistant without registration. Users can generate educational responses and diagrams instantly; however, conversation history is not stored.

Some of the key features implemented in the system include:

- Context-aware conversational AI that maintains conversation history for coherent multi-turn interactions.
- Automatic Mermaid diagram generation to visually represent concepts.
- Structured educational responses with fixed explanatory points for clarity.
- JWT-based authentication and authorization with access and refresh tokens.
- Guest access mode for system demonstration without data persistence.

System Design

MongoDB Schema Design

The database was designed using MongoDB, with schemas defined using Mongoose to ensure consistency and maintainability. Although MongoDB is schema-less by nature, Mongoose was used to enforce structured data models and improve data integrity.

A combination of embedded documents and referenced collections was used to ensure scalability and efficient data relationships. This design approach simplified API development and ensured predictable data handling across the application.

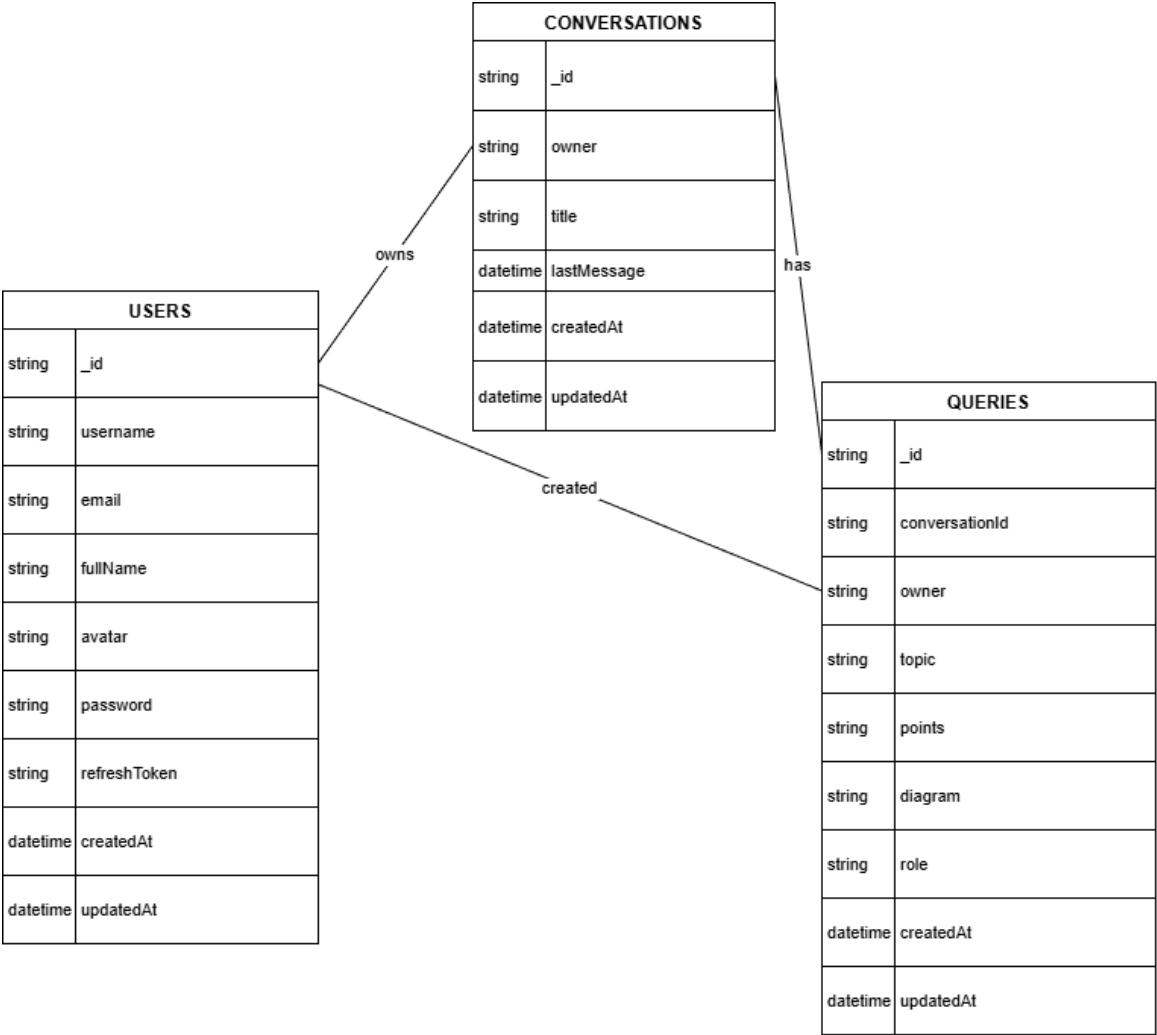


Figure 3.1: Illustrates of MongoDB schema and relationships

Sequence Diagram for Conversational Flow

The sequence diagram illustrates the interaction flow when a user submits a query. The frontend sends the user's question to the backend, which validates the request, constructs conversation history, and forwards it to the AI service. The AI generates structured educational content and a Mermaid diagram, which is stored in the database and returned to the frontend for rendering.

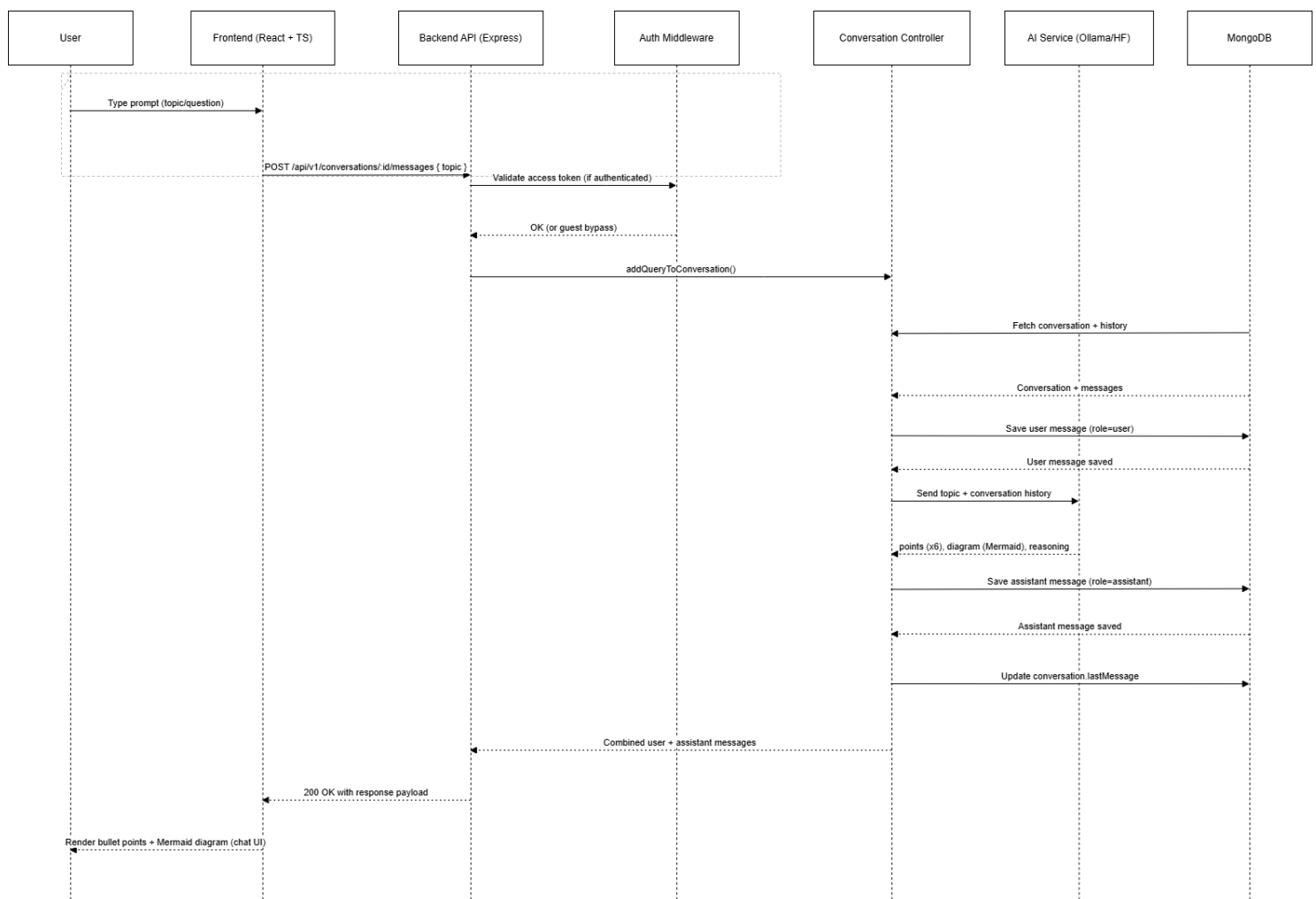


Figure 3.2: Sequence Diagram

Use Case Diagram

The use case diagram presents a high-level view of system interactions between users and the AI assistant. It highlights the main functionalities such as asking questions, generating diagrams, managing conversations, and accessing learning history.

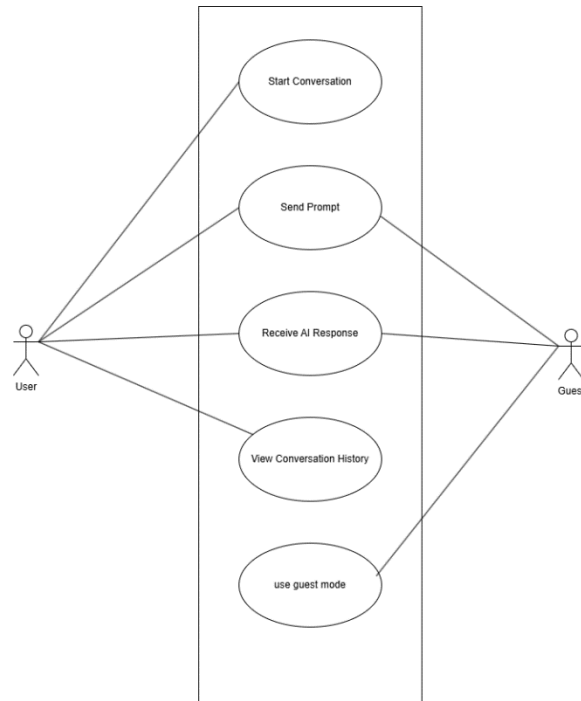


Figure 3.3: Use Case Diagram

3.3.1 Tools Used

The tools and technologies used during the development of the system are described below.

Frontend Technologies

React JS with TypeScript

React JS was used as the core frontend framework to build an interactive, component-based user interface. TypeScript was integrated to provide static type checking and improve code quality.

Reusable components were developed for chat interfaces, conversation lists, and diagram rendering. React hooks such as `useState`, `useEffect`, and `useRef` were used to manage component state, lifecycle events, and dynamic UI behavior.

TanStack Query (React Query)

TanStack Query was used to manage server-side state and asynchronous data fetching. It provided built-in caching, loading states, and error handling, reducing the need for manual state management.

Mutations were used for creating conversations and messages, while query invalidation ensured real-time updates across the application.

Backend Technologies

Node.js

Node.js was used to handle server-side logic, request processing, and AI integration workflows. It enabled efficient handling of asynchronous operations and RESTful API development.

Express.js

Express.js was used to define API routes, middleware, and authentication logic. It facilitated structured backend development and seamless communication between frontend and database services.

MongoDB

MongoDB served as the primary NoSQL database, storing user information, conversation metadata, and message history. Its flexible document-based structure supported scalable conversational data storage.

Other Tools

Ollama/HuggingFace

Ollama was used to run local large language models for generating AI responses and diagrams. HuggingFace APIs were optionally integrated to support additional AI models and experimentation.

Git and GitHub

Git and GitHub were used for version control and project collaboration. They enabled tracking of code changes, feature branching, and overall project management.

Postman

Postman was used to test and validate backend API endpoints during development, ensuring correct request handling and response structures.

3.4 Tasks / Activities Performed

3.4.1 Feature Development

Actively contributed to both frontend and backend development of the AI learning assistant.

Tasks Performed:

- Developed the conversational chat interface using React and TypeScript.
- Implemented conversation creation, deletion, and management features.
- Designed backend APIs for handling conversations and AI-generated responses.
- Integrated Mermaid diagram rendering within the frontend UI.

3.4.2 AI Integration

Integrated AI services to generate educational responses and diagrams.

Tasks Performed:

- Designed structured prompts for AI-generated educational output.
- Integrated Ollama for local AI inference.
- Implemented backend logic to process AI responses and store results.

3.4.3 Backend and API Integration

Developed RESTful APIs to support user authentication, conversation handling, and message storage.

Tasks Performed:

- Built APIs using Node.js and Express.
- Integrated frontend with backend services using Axios and React Query.
- Ensured secure access using JWT-based authentication.

3.4.4 Optimization

Focused on improving performance and maintainability.

Tasks Performed:

- Optimized database queries for conversation and message retrieval.
- Implemented caching strategies using React Query.
- Improved frontend rendering performance for dynamic diagrams.

Chapter 4: Conclusion and Learning Outcome

4.1 Conclusion

My internship at Softech Foundation Pvt. Ltd. was a highly rewarding experience that successfully bridged the gap between academic knowledge and real-world software development practices. Throughout the internship, I was actively involved in the design and development of an AI-powered full-stack learning platform, which provided me with practical exposure to the complete software development lifecycle.

Working on the GraphGeneration – AI-Powered Learning Assistant project significantly strengthened my understanding of modern full-stack development using the MERN stack. I gained hands-on experience in developing scalable backend APIs, designing structured database schemas, and integrating AI-driven functionalities into a web-based system. The project also enhanced my understanding of how conversational AI systems can be designed to support educational use cases through structured responses and visual diagram generation.

In addition to technical growth, the internship allowed me to work in a collaborative team environment where regular discussions, code reviews, and feedback sessions were conducted. This experience improved my communication skills, problem-solving approach, and ability to work effectively within a professional development team. Overall, the internship played a crucial role in strengthening both my technical and professional competencies.

4.2 Learning Outcome

During my internship at Softech Foundation Pvt. Ltd., I gained substantial hands-on experience in full-stack web application development using the MERN stack, which helped me solidify my understanding of frontend, backend, and database integration. I became proficient in developing interactive user interfaces using React and TypeScript, managing server-side logic with Node.js and Express.js, and designing scalable data models using MongoDB.

Through the implementation of AI-driven features, I developed a clear understanding of how large language models can be integrated into real-world applications. I learned how to structure AI prompts, process AI-generated responses, and present them in a user-friendly

format, including visual representations such as Mermaid diagrams. This experience enhanced my understanding of applied artificial intelligence in educational systems.

The internship also strengthened my problem-solving abilities through continuous debugging, testing, and optimization of system components. I gained experience using tools such as Git and GitHub for version control and collaboration, as well as Postman for API testing and validation. Regular interaction with mentors and team members helped me improve my communication skills, teamwork, and ability to accept and apply constructive feedback. These learning outcomes have significantly contributed to my professional growth and prepared me for future opportunities in software development.

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Appendices

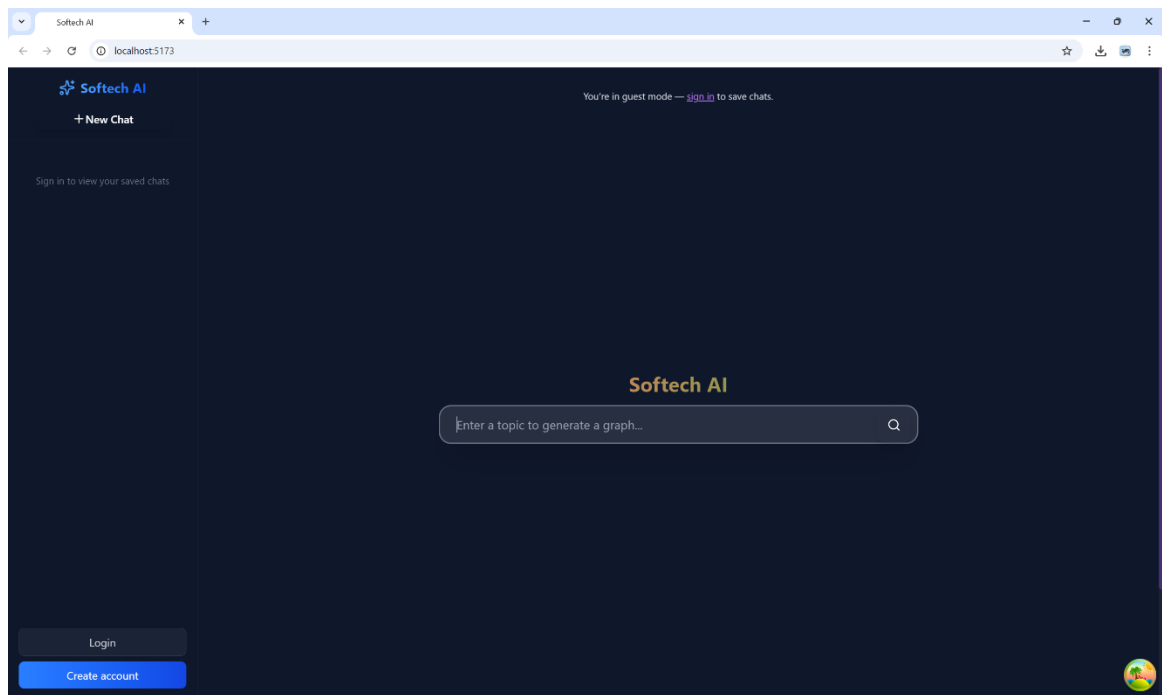


Figure: Landing Page

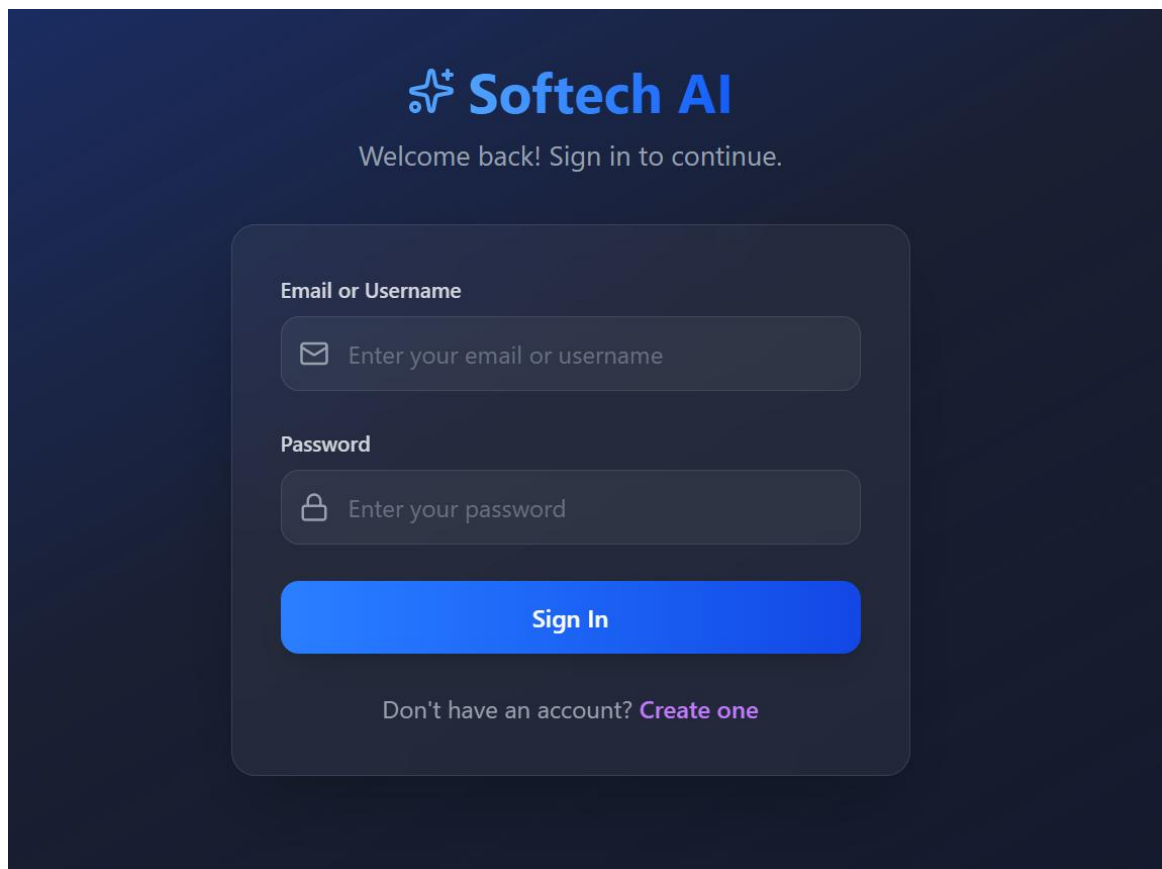


Figure: Login Page



Create your account to get started.

Username

 Choose a username

Full Name

 Enter your full name

Email

 Enter your email

Password

 Create a password

Confirm Password

 Confirm your password

Create Account

Already have an account? [Sign in](#)

Figure: Registration Page

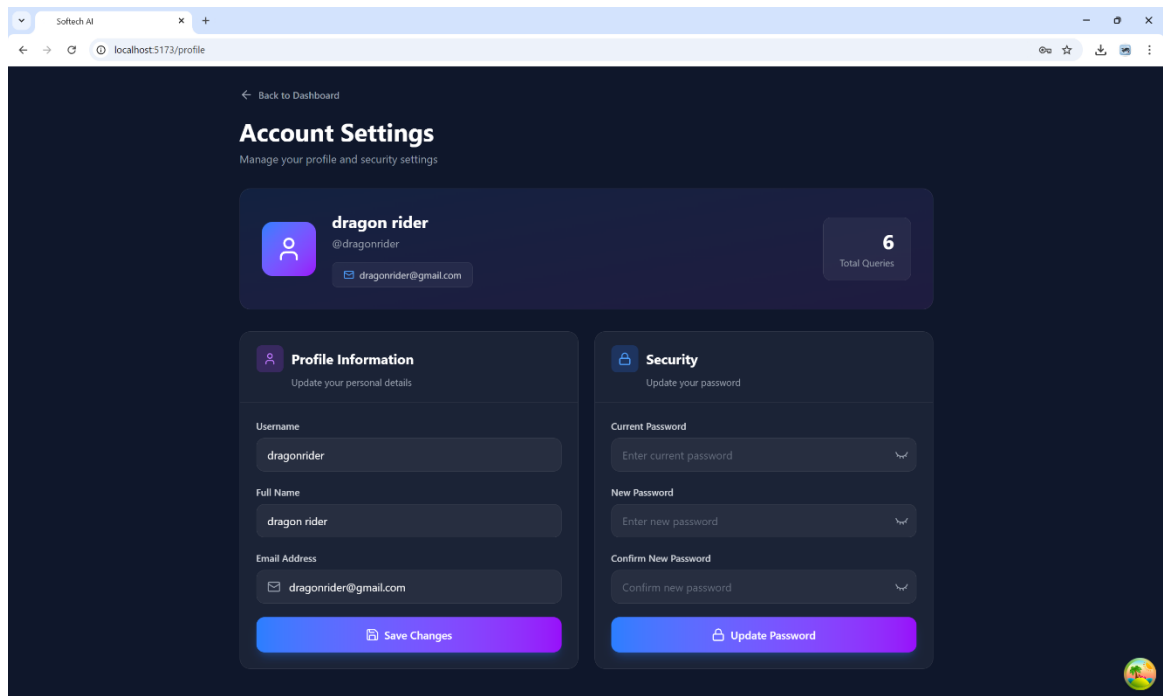


Figure: Account Setting Page

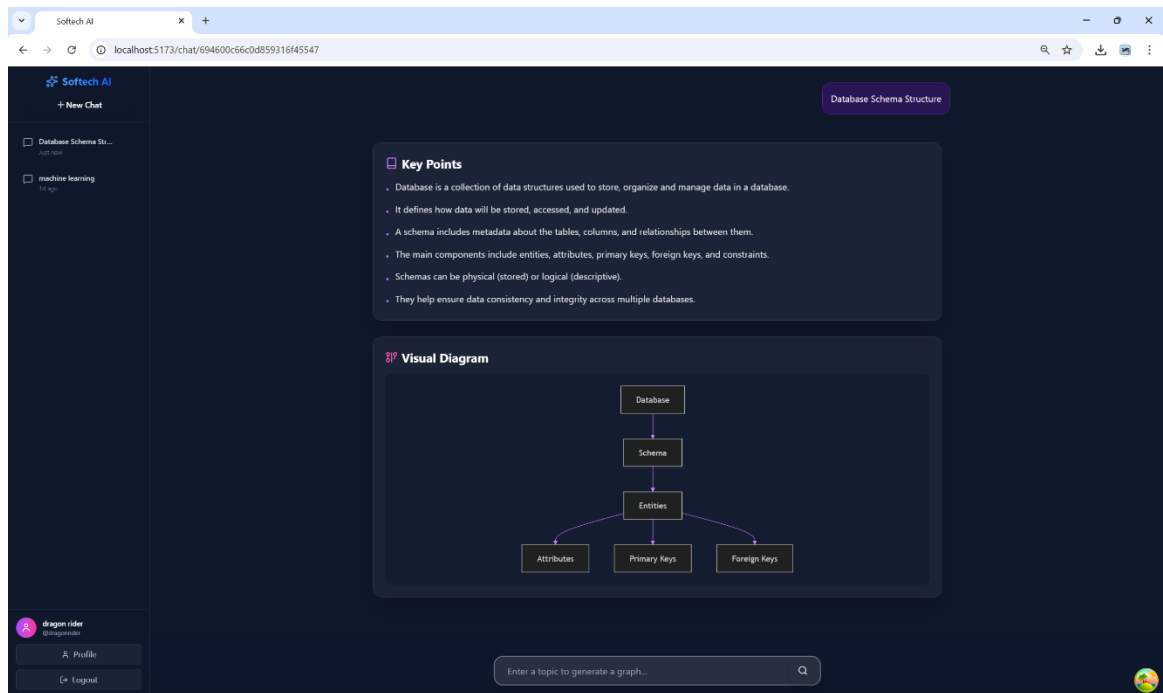


Figure: Creating Diagram